

THE GREAT AGENT HACKATHON 2026 · FINAL ROUND

CODEMIX SKILL

The safety + decision layer between AI agents and their actions

We put an intent-validation safety gate between what the voice agent hears and what the agent is allowed to do.

Initiative: Codemix Skill · Owner: Team Ramanathan & Sadhana · Track 1: Customer & Employee Experience

THE GAP

Voice agents can immediately call a refund, cancellation or replacement API. One missed negation turns the right conversation into the wrong transaction.

OUR PRODUCT

A plug-in between the AI agent and its API/tool calls. It validates intent, negation, confidence, sentiment and SOP rules before execution.

FRESHWORKS EXTENSION

Unsafe, uncertain or escalated calls become Freshdesk tickets with transcript, decision context and action details for a human agent.

Understand

Decide

Act

Audit

*“Order cancel **mat** karo, bas address change karna hai.”*

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Problem Statement

In code-mixed voice, one short word can reverse the meaning of an API action

HINDI EXAMPLE

*“Order cancel **mat** karo, bas address change karna hai.”*

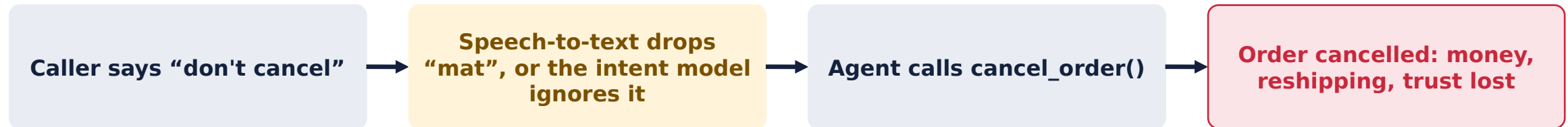
Correct intent: do NOT cancel → update address

TAMIL EXAMPLE

*“Refund **vendam**, replacement anuppunga.”*

Correct intent: do NOT refund → send replacement

HOW IT BREAKS TODAY



48%

of negated requests would still trigger a refund / cancel in our own Round-1 engine

Measured: 11 of 23 team-written test sentences (Hindi, Tamil, English, Telugu, Kannada, Bengali)

26%

of cash-on-delivery orders in India return to origin, so refund / cancel / return calls are high-volume

Source: Shipway ShipNotes report, FY25 (Jul 2025)

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Why Now & Competitive Landscape

AI agents have write access, and nobody gates what they are allowed to do on a misheard word

WHY NOW



AI agents can act

Tool calling and MCP let agents trigger refunds, cancellations and returns directly.



Code-mixing is normal

Indian callers switch language mid-sentence; the risk sits in small intent-flipping words.



The voice stack is ready

VoBiz, Sarvam, Claude and ElevenLabs make a checked, real-time voice agent possible now.

27-70%

Hinglish word error rate across speech-to-text models. Speech recognition alone can't be trusted with irreversible actions. *Deepgram*

Our timing Not another voice bot: the layer that decides when an agent may call an API.

COMPETITIVE LANDSCAPE

Alternative

Strength

Gap

Do nothing

Zero effort

Risky actions run on the first interpretation

Confirm-everything IVR

Safe confirmation

Friction on every action, even low-risk ones

Generic LLM guardrails

Policy, toxicity, PII

Not an action gate for code-mixed negation + speech confidence

Codemix Skill

Code-mixed intent → risk check → ALLOW / CONFIRM / BLOCK

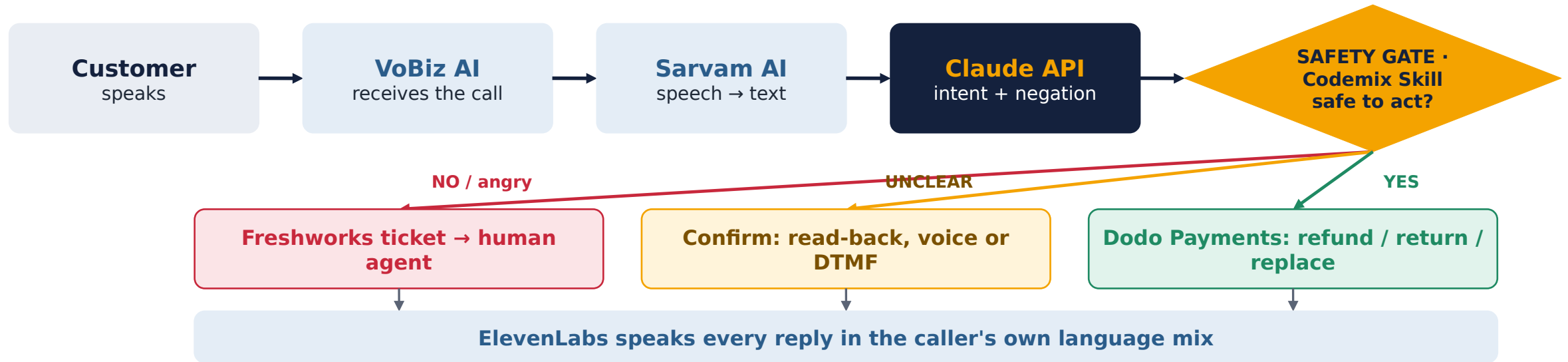
Unsafe cases go to Freshworks with full context

Our edge Risk-based action gate + human escalation + a full audit record.

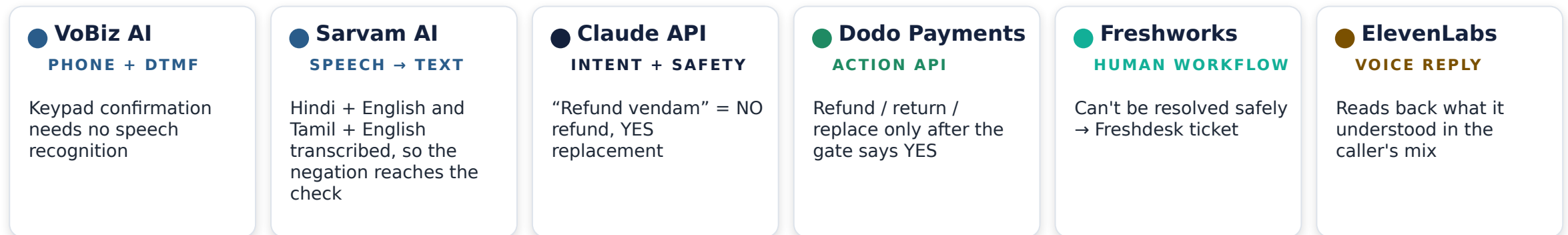
5

Solution: the Intent-Validation Safety Gate

Between what the voice agent hears and what it is allowed to do; each part has one job in preventing a wrong action

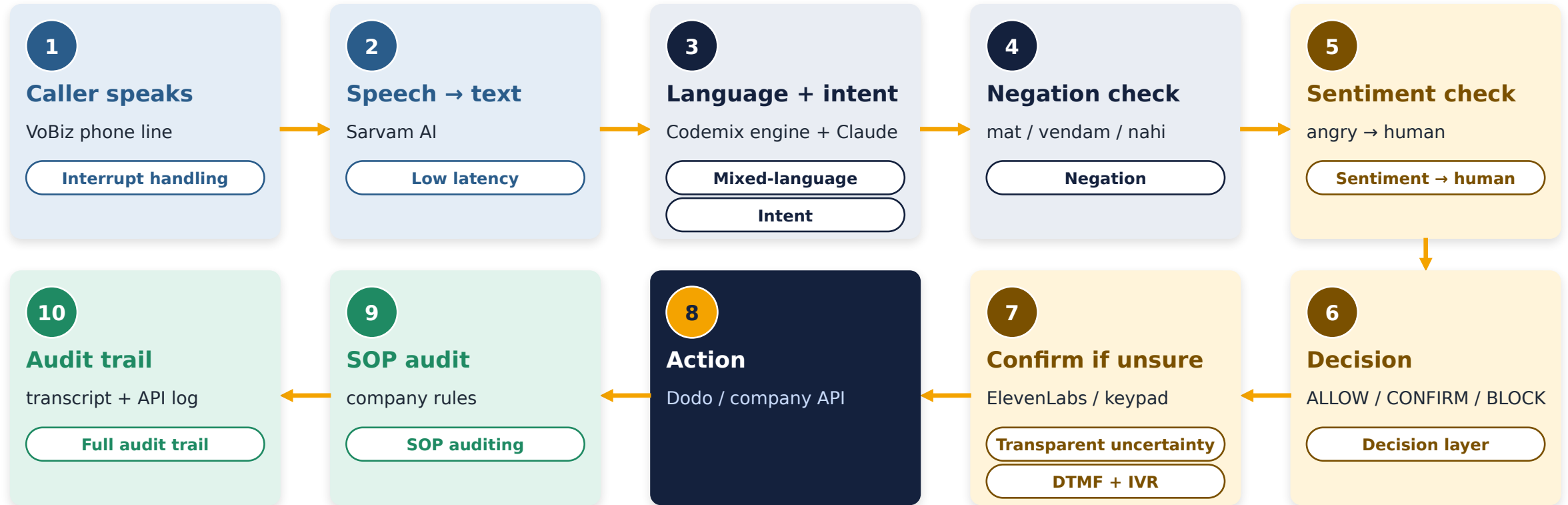


HOW EACH PART PREVENTS A WRONG ACTION



5 One Call, End to End — All 11 Features

Every feature appears only where it adds value; routine calls take the fast path



Sentiment: Claude scores caller sentiment every turn; two negative turns or an explicit request for a human → Freshdesk ticket.

Low latency: fast offline checks first (1.87 ms measured); Claude runs only before sensitive actions.

SOP audit: every call is checked against company rules (authorisation, required confirmation, escalation) → PASS / VIOLATION + audit reference; transcript, decisions and API calls are stored.

5

The Safety Decision & Transparent Confirmation

The agent never silently guesses on an irreversible action



TRANSPARENT CONFIRMATION IN PRACTICE

CALLER

"Refund... vendam... maybe replacement."

AGENT • ElevenLabs, caller's mix

"I'm not fully sure whether you want a refund or a replacement. Press 1 for replacement, 2 for a human agent."

CALLER • keypad via VoBiz

presses 1 → no speech-to-text ambiguity

Interrupts: VoBiz voice-activity detection stops ElevenLabs playback the moment the caller speaks; the new words become the latest input.

BLOCK / ROUTE

Negation conflicts with the action, or confirmation fails → stop API + Freshdesk.

EXAMPLE OUTCOME

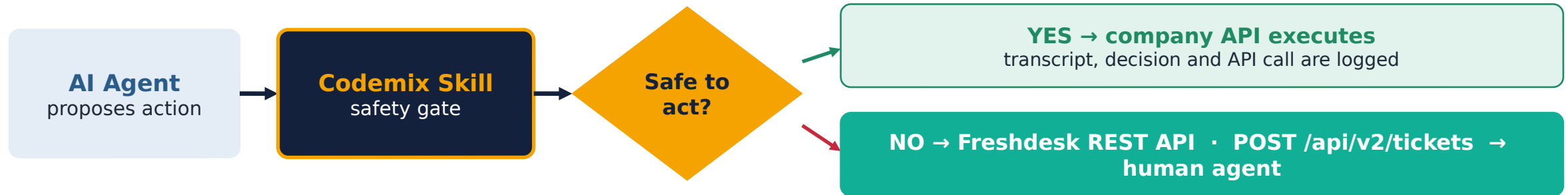
"Refund vendam, replacement anuppunga."

BLOCK refund → confirm replacement → if unclear, Freshdesk ticket

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Freshworks Integration — Human-in-the-Loop

Freshdesk takes the handoff; also works with Freshcaller as the voice channel and alongside Freddy AI Agent



A ticket is created when...

- the action is unsafe or unclear
- the caller doesn't confirm
- the caller is angry / distressed
- the AI should not guess



What we add to Freshdesk

Code-mixed intent validation, action gating, transparent confirmation and structured escalation, on top of ticketing, agent workspace and customer history.

What the human agent receives in the Freshdesk ticket

CONVERSATION

Call ID + timestamp

Original code-mixed transcript

Detected languages

English summary

Voice / DTMF source

DECISION

Intent + confidence

Negation detected

Safety verdict

Requested API action

Confirmation status

OPERATIONAL

Priority / tags

SOP result

Reason for escalation

Attempted API call

Audit reference

5

Agentic Design, MVP Scope & Early Proof

A plug-in control point that intercepts the proposed action before the company API is called

AGENTIC DESIGN: THE PLUG-IN CONTROL POINT



BUILT TODAY

Code-mix engine + intent scoring · MCP server · live Freshdesk ticket API · negation gate v0 + benchmark · Vercel demo

MVP (3 MONTHS)

VoBiz + Sarvam + ElevenLabs voice path · Claude safety gate · Dodo action path · Freshdesk escalation · Hindi + Tamil + English · audit trail

NEXT

Sentiment + interrupt tuning · SOP rule editor + reports · more Indian languages · Freshworks Marketplace app

EARLY PROOF (MEASURED)

Wrong refund / cancel on negated requests

47.8%

0.0%

Negated requests

■ Round-1 engine alone ■ With negation gate v0

11 of 23 → 0 of 23 wrong actions; 0 of 20 genuine requests blocked. 43 team-written sentences in 6 languages; the pilot measures real calls.

6-9 Customers, Value, Pricing & Go-to-Market

Start where AI agents already have permission to act; pilot → proof → Freshworks distribution

6 Target Customers

Primary: D2C, e-commerce and fintech brands on Freshdesk deploying AI voice agents for Indian customers.

Secondary: BPOs needing one safety / policy layer across many clients.

Protected actions: refunds, returns, replacements, cancellations.

Why now: ~26% of COD orders return to origin (Shipway FY25), so these calls are high-volume.

7 Value Proposition

Plug it in once: the existing agent serves mixed-language callers, and nothing irreversible happens on a misheard word.

Without

- Acts on first guess
- Dropped “mat” = wrong refund
- Blind escalation, no record

With Codemix Skill

- Checks intent + negation first
- Confirms when unsure
- Freshdesk handoff + audit trail

8 Pricing & Packaging (proposed)

Free Monitor mode: flags risky actions, never blocks

\$0.03 per guarded action: plug-in on the customer's own agent

₹8 / min full voice agent: VoBiz + Sarvam + Claude + ElevenLabs + Freshdesk

Custom Enterprise: SOP rule packs, India data residency, SLA

Priced above run cost (~55% gross margin, slide 10). One prevented ₹1,000 wrong refund pays for ~390 guarded actions.

9 Go-to-Market Plan

- 1 3 Indian D2C design partners on Freshdesk
- 2 Free monitor-mode pilot
- 3 Publish wrong-action proof + demo video
- 4 Freshworks Marketplace listing
- 5 Scale via Freshdesk deals + BPO partners

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Costs, ROI, Success Metrics & Risks

What it costs to build and run, what it returns, how we measure it, and what could go wrong

10 Costs & Unit Economics

~\$90K

3-month MVP build: M1 data · M2 gate + voice · M3 Freshdesk + pilot

Engineers \$60K · linguist \$10K · PM \$8K · APIs \$7K · buffer \$5K

RUN COST PER 3-MIN VOICE CALL

VoBiz telephony (₹0.45/min)	₹1.35
Sarvam STT (₹30/hr)	₹1.50
ElevenLabs Flash (\$0.05/1K chars)	₹6.40
Claude checks (Haiku 4.5 / Sonnet 5)	₹0.71
Hosting + transcript storage	₹0.43
Total ≈ ₹3.5/min vs ₹8 price	₹10.4

Plug-in only: ₹1.1 cost vs ₹2.55 price per action. Dodo and Freshdesk fees stay on the customer's own plans.

11 Success Metrics

	TODAY	TARGET
Wrong actions on negation	48%*	<0.5%
Engine latency	1.87ms†	<5ms
p95 reply (sensitive)	—	<900ms
Confirmation rate	—	≤15%
Audit coverage	—	100%
Reversal tickets	—	−30%
Adoption (2 qtrs)	—	15%

** Round-1 engine, no gate, 23-sentence test set † npm test average*

12 Risks & Assumptions

STT drops the negation

→ confirm every risky action

Safety check adds latency

→ deep checks only when sensitive

Dialect / spelling variation

→ lexicon + model fallback

Read-back friction

→ ≤15% confirmation budget

ASSUMPTIONS

- Pilot brands expose refund / return APIs
- Callers accept a short confirmation
- Sarvam is accurate enough with the fallback

RETURN ON INVESTMENT

illustrative · full assumptions in speaker notes · validated in the pilot

~3.8×

Customer ROI

₹1.9L saved vs ₹51K fee / month

0.26%

Break-even

of guarded actions going wrong

\$349K

Year-1 revenue

vs \$90K build · ~\$195K gross profit

~8 mo

Payback

after launch, on gross profit